Arunabha Majumder

Mechanical Design Engineer | SolidWorks, Electromechanical Assemblies, Prototyping and Testing

+45 71 62 24 00 | Aalborg, Denmark | somrkmv1997@gmail.com | 97arunabhasit027@gmail.com
Portfolio: arunabha22.github.io/arunabha-majumder | LinkedIn: linkedin.com/in/arunabha-majumder-681264107

SUMMARY

Mechanical design engineer with a PhD in progress in Mechatronics and Robotics at Aalborg University, plus an MTech in Mechatronics and a BTech in Mechanical Engineering. I design electromechanical assemblies in SolidWorks, build and test the prototypes and test rigs myself, and find and fix what goes wrong. Best Research Paper Award, IFToMM ISRM 2026. Certified SolidWorks Associate.

EXPERIENCE

PhD Researcher, Mechatronics and Robotics (project VIEXO) | 2023 - Present
Aalborg University, Department of Materials and Production, Aalborg, Denmark

- Designed a hybrid shoulder exoskeleton in SolidWorks, where a spring carries most of the load and a small 6 Nm motor adds the rest; it cut energy use by more than 25% (Best Paper Award, IFToMM ISRM 2026).
- Took the assembly from hand sketches to 3D CAD models and drawings, then built and rebuilt prototypes with 3D printing, improving weak parts after each test round.
- Integrated the motor, encoder and force sensors into the mechanical frame and built my own test rigs; found and fixed faults in mounting, wiring, sensors and data logging.
- Validated a force estimate on a 5-bar parallel robot (variable-stiffness actuators, robot Jacobian) against a reference force sensor, within 10% RMS error.
- Worked in a team of 3 on an Assist-as-Needed controller (my part: equations such as the Fourier fit, and GUI behaviour); collected data from 12 participants; presented at international conferences and taught and supervised students (2023 - 2025).

MTech Researcher, Pneumatic Artificial Muscles (Master's thesis) | 2020 - 2022
CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

- Fabricated and assembled pneumatic artificial muscles to an existing lab design and ran experiments on the lab's setup with valves, pressure sensors and DAQ.
- Collected the data, fed it into the lab's existing neural network and used its output to design the outer PID loop of a gain-scheduled controller (0.3-0.78% error, IEEE CONECCT 2022).

Bachelor's Capstone, Upper-Limb Rehabilitation Exoskeleton | 2018 - 2019
Siddaganga Institute of Technology, Tumakuru, India

- Designed, machined, assembled and user-tested the device. Best Major Project Award and an INR 100,000 TATA Technologies grant.

SKILLS

- **CAD:** SolidWorks (CSWA): parts, assemblies and technical drawings
- **Systems:** Electromechanical integration: motors, encoders, force/torque and pressure sensors, DAQ
- **Prototyping:** 3D printing, test rigs; machining and assembly (Bachelor's)
- **Analysis and software:** MATLAB/Simulink, Python, LabVIEW (basic)
- **Drawing and communication:** Diploma in Fine Arts; concept sketches, technical illustrations, clear figures for reports and talks
- **Languages:** English (C1, fluent), Bengali (native), Hindi (conversational)

EDUCATION

PhD, Mechatronics and Robotics (defence expected in 2027) | 2023 - Present
Aalborg University, Denmark

MTech, Mechatronics | 2020 - 2022
AcSIR - CSIR-CMERI, India

BTech, Mechanical Engineering | 2015 - 2019
Siddaganga Institute of Technology (VTU), India

AWARDS AND PUBLICATIONS

- Best Research Paper Award, IFToMM ISRM 2026; Certified SolidWorks Associate (CSWA)
- Majumder, A., et al. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." ISRM 2026, Springer.
- Majumder, A., et al. "Neural Network-Based Gain Scheduled Position Control of a Pneumatic Artificial Muscle." IEEE CONECCT, 2022.

REFERENCES

Available upon request.